

Final
CSED501

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Instructions: Complete the questions below and upload a PDF containing your written answers to Gradescope.

- If using this provided pdf, keep your answers within the provided space.
 - Answers can be provided in any PDF format, but answer locations should be tagged when uploading to Gradescope.
 - Show your work; this makes giving partial credit easier!
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You are given a dataset containing 5,000 rows and 24 inputs describing customers of a retail company. The dataset includes:

- **Numerical inputs:** age, income, number_of_visits, avg_purchase_value, and others.
- **Categorical inputs:** city, membership_tier, preferred_device, and others.
- **Target 1:** A binary outcome variable `churn` (0/1).
- **Target 2:** A continuous outcome variable `lifetime_value`: the total expected revenue or profit that a customer will generate for the company over the entire duration of their relationship with the business.

The dataset contains mixed input scales. Answer the following three (3) questions about this given dataset. **Each question is separate: the information in each does not carry over to the next.**

Problem 1 **12 pts**: Assume you are using this dataset to train a machine learning algorithm that cannot handle categorical data, and has hyperparameters that you need to select. You would also like to predict performance on future data. Describe the preprocessing steps you would take to prepare this dataset and provide an order in which to execute these steps.

Answer:

The pre-processing steps I would take to prepare the data are,

1. Pick a target to predict and then drop the other from the dataset.
2. Split the data into training, test and validation. Specifically a 70/15/15 split.
3. Analyze the target in the training set for outliers(value) or class imbalance (churn) and use appropriate techniques on the dataset and model comparison.
4. Analyze the features of the training set for missing values and act as appropriate. (e.g. drop sample, impute with numerical with median)
5. Encode categorical features with one-hot encoding
6. Scale numerical features.
7. Use the fitted transformer (e.g. scaling data) on the validation on test set features.

At this point the data would be ready to be run through the identified models and associated hyperparameters for the problem use something like GridSearchCV.

Problem 2 **18 pts**: Assume you are using linear regression on this dataset to predict the continuous outcome variable, `lifetime_value`. Answer the following questions in 1-3 sentences.

- (a) Linear regression is sensitive to features with different scales. Explain why feature standardization can improve model stability and interpretability.
- (b) Linear regression assumes a linear relationship between each feature and the target. Describe one way to check whether a feature has a linear relationship with the target.
- (c) Linear regression is sensitive to outliers. Describe one method of detecting outliers.

Answer:

a) Feature standardization can improve model stability and interpretability by scaling the inputs so that the weights determined by the normal equations don't compound errors in the output and gradient descent converges more easily. Interpretability is improved because standardization allows us to compare their relative contribution of the weights to the output.

b) One way to check whether there's a linear relationship between each feature and the target is to use a scatter plot and visually inspect whether the points lie along a line.

c) For outlier detection I would use one-class SVM since we are dealing with a decent number of features/dimensions. Many other models we've learned, like clustering, struggle with higher dimensions or are themselves sensitive to outliers.

Problem 3 **18 pts**: You are comparing the performance of decision trees to logistic regression on this dataset to predict churn (binary classification). Answer the following questions in 1-3 sentences.

- (a) Do decision-tree-based models generally require feature standardization? Explain why or why not.
- (b) You decide to optimize the maximum depth hyperparameter for your decision tree model, and are choosing between k -fold cross-validation and a single train/validation/test split. How does k -fold cross-validation improve model evaluation compared to using a single train/test split? What is one potential downside of using k -fold cross-validation?
- (c) For your logistic regression model, you use all inputs as features and create some higher-dimensional features ($d = 8$) in addition. When examining the model's performance on the validation set, you observe high **variance**. Identify at least one algorithm/model that we learned in class that you could use to lower the variance, and briefly explain how using this algorithm/model lowers variance (an intuitive explanation will suffice; no math is needed in this answer). When variance is decreased, is the model's **bias** on the validation set most likely to decrease, increase, or stay the same?

Answer:

- a) No, because the splits in a tree are determined by a threshold per feature and so multiplying by a constant doesn't change the decision at a node.
- b) K-fold cross-validation improves model evaluation compared to a single split by reducing variance through averaging the performance metric. The potential downside to K-fold cross-validation is the increased compute requirements.
- c) One can use regularization to lower variance via the Lasso or Ridge algorithms. Those algorithms shrink the coefficients so that have a large impact on the prediction, which in effect lowers variance. Decreased variance is most likely to increase bias because of the bias-variance tradeoff.

In this class, you have learned the following algorithms/models/approaches:

- Linear Regression
- Gradient Descent
- L1 Regularization
- L2 Regularization
- Principal Component Analysis (PCA)
- Perceptrons
- Support Vector Machines (SVMs)
- Logistic Regression
- Decision trees
- Random Forests
- AdaBoost
- K-nearest neighbors (k-NN)
- K-means
- Gaussian Mixture Models (GMMs)
- Agglomerative clustering
- DBSCAN
- Neural Networks
- Markov Models
- Value Iteration
- Q-Learning

The following four (4) problems will pose a problem statement. **For each problem, provide two (2) algorithms/models/approaches from the above list:** the **best** one (the one that fits the proposed problem specs the best), and a second that **could solve the proposed problem but may not be as optimal or fit all of the specs**. For the second algorithm/model/approach, describe in 1-2 sentences at least one reason it is not optimal/doesn't fit all specs.

Problem 4 **12 pts**: A warehouse robot must navigate a known grid layout with obstacles to reach a designated area. Engineers have:

- A complete description of how the robot moves, including slippage and turn-error probabilities.
- A fully specified reward function.
- A relatively small, discrete environment.
- A desire to compute a policy that is guaranteed to be optimal.
- A full model of the environment.

As described previously, provide two (2) algorithms/models/approaches from the list.

Answer:

Best: Value Iteration because there is a complete specification of the environment and require a optimal policy.

Possible: Q-Learning because one could solve the problem, but are not guaranteed an

Problem 5 **12 pts**: A maritime analytics group is studying GPS traces of ships traveling near a coastline to identify common shipping routes and areas where vessels frequently travel together. The data exhibits the following characteristics:

- Long, irregular patterns corresponding to typical vessel paths.
- A substantial amount of noisy, scattered GPS points from faulty readings.
- An unknown number of meaningful route patterns.

As described previously, provide two (2) algorithms/models/approaches from the list.

Answer:

Best: The problem is not a prediction one, but rather we want to explore grouping data points together that could routes. A clustering algorithm is what we want to use. In this specific case DBSCAN is designed to identify an unknown number of groups, irregular shapes and outliers. Core points identify routes and outliers would be noise points. These characteristics fit the description of our problem perfectly.

Possible: Gaussian Mixture Models could loosely group points together with its elliptical shapes, but requires specifying the number of routes/groups up front and is more sensitive to noise.

Problem 6 **12 pts**: A retail analytics company is studying how various store metrics—such as weekly advertising spend, number of staff hours, average customer foot traffic, number of promotions run, and store size—affect the weekly revenue at each location.

Exploratory analysis reveals that:

- The relationship between each metric and revenue is approximately linear.
- The dataset includes tens of thousands of store-weeks, with mostly numerical features.
- Management requires a model that provides clear insight into how each factor contributes to revenue.

As described previously, provide two (2) algorithms/models/approaches from the list.

Answer:

Best: Linear Regression. The relationship between each feature and the target is described as linear and the coefficients can be interpreted for their contribution to revenue. Also, the normal equation for linear regression can be computed efficiently and so can handle the largish dataset.

Possible: Random Forest. The model could be used to predict revenue accurately. However it doesn't offer a clear way to see how each feature contributes because the thresholding in the splits don't offer the same clear relationship.

Problem 7 **12 pts**: A manufacturing quality-control team is working with a dataset in which each product is represented by a very high-dimensional set of sensor measurements—over 30,000 recorded features per item. Their goal is to separate items into two groups: those that pass inspection and those that fail.

During exploratory analysis and discussions on deployment, they find that:

- The two groups are almost entirely linearly separable
- Models must make predictions quickly once deployed.
- Model interpretability is not important for this task.
- As much as possible, the model should generalize well to new data not in the original dataset.

As described previously, provide two (2) algorithms/models/approaches from the list.

Answer:

Best: SVM. This model is designed for high-dimensional and linearly separable data, which fits our problem statement. Additionally, SVM can produce a lower variance model on our dataset by maximizing the margin, which provides the most distance between the predicted hyperplane and points of the two groups. This means new points are less likely to be misclassified.

Possible: Logistic Regression, maybe with regularization to reduce features. While this model works well with linearly separable data and can do the prediction task it's not going to generalize as well because the possible hyperplanes found via the data doesn't maximize the margin like SVM does.

